

Data Handling and Visualization

Class Activity

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Date: 24.07.2026

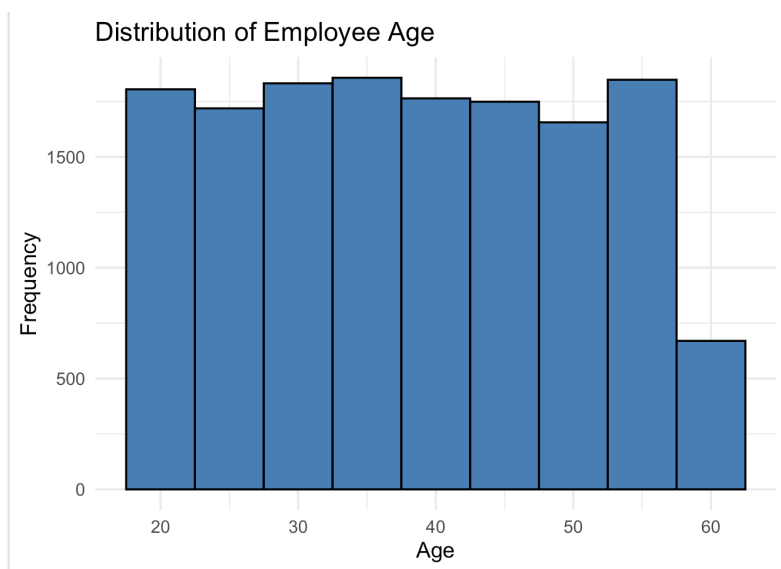
Kaggle Dataset: Employee Attrition Classification Dataset

1. Histogram (Employee Age)

```
library(ggplot2)

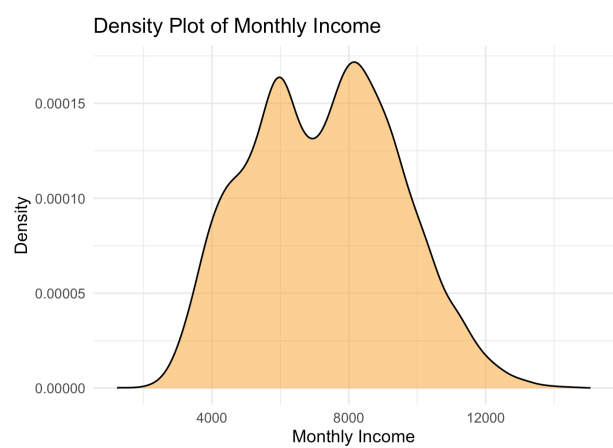
employee <- read.csv("~/Downloads/employee.csv")

ggplot(employee, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "steelblue", color = "black") +
  labs(title = "Distribution of Employee Age",
       x = "Age",
       y = "Frequency") +
  theme_minimal(base_size = 14)
```



2. Density Plot (Monthly Income)

```
R
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
ggplot(employee, aes(x = MonthlyIncome)) +
  geom_density(fill = "orange", alpha = 0.6) +
  labs(title = "Density Plot of Monthly Income",
       x = "Monthly Income",
       y = "Density") +
  theme_minimal(base_size = 14)
```



3. Box Plot (Monthly Income)

```
R
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
ggplot(employee, aes(y = MonthlyIncome)) +
  geom_boxplot(fill = "skyblue") +
  labs(title = "Box Plot of Monthly Income",
       y = "Monthly Income") +
  theme_minimal(base_size = 14)
```



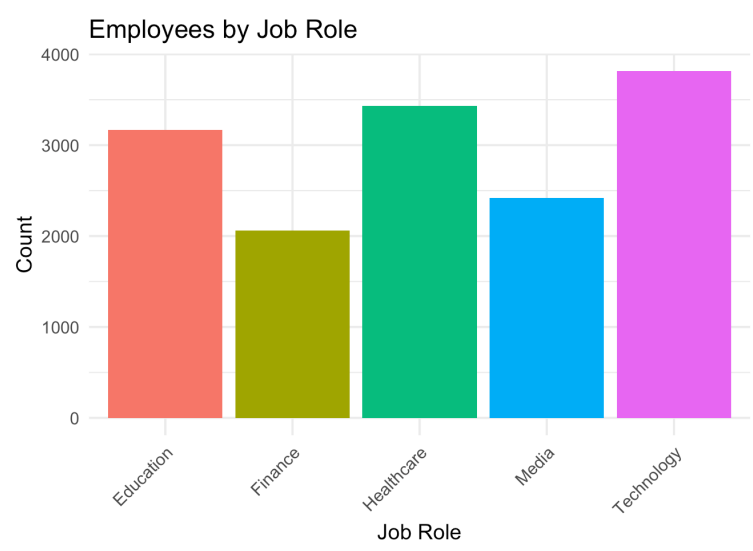
4. Violin Plot (Monthly Income by Gender)

```
R
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
ggplot(employee, aes(x = Gender, y = MonthlyIncome, fill = Gender)) +
  geom_violin(trim = FALSE) +
  labs(title = "Monthly Income by Gender",
       x = "Gender",
       y = "Monthly Income") +
  theme_minimal(base_size = 14)
```



5. Bar Chart (Job Role)

```
R
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
ggplot(employee, aes(x = JobRole, fill = JobRole)) +
  geom_bar() +
  labs(title = "Employees by Job Role",
       x = "Job Role",
       y = "Count") +
  theme_minimal(base_size = 14) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```



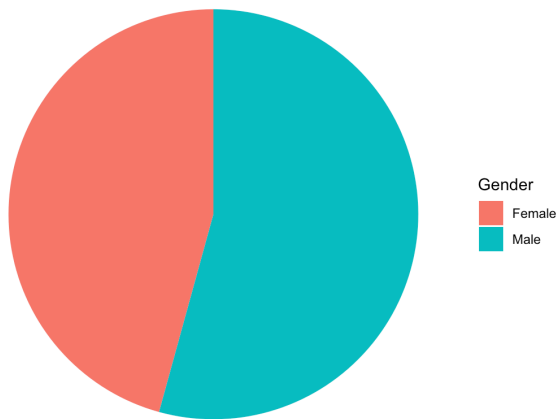
6. Pie Chart (Gender Distribution)

```
R
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
gender_count <- as.data.frame(table(employee$Gender))
colnames(gender_count) <- c("Gender", "Count")

ggplot(gender_count, aes(x = "", y = Count, fill = Gender)) +
```

```
geom_col(width = 1) +
coord_polar(theta = "y") +
labs(title = "Gender Distribution") +
theme_void()
```

Gender Distribution

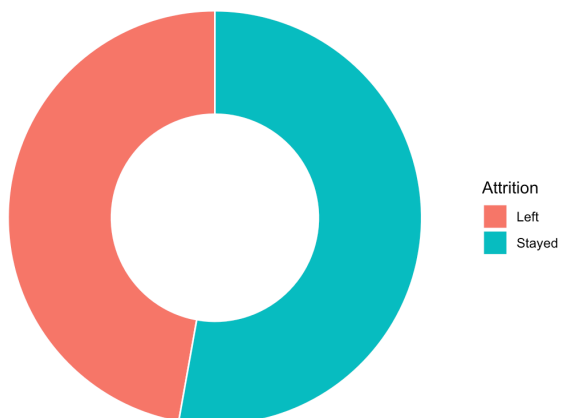


7. Donut Chart (Attrition)

R

```
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
attrition_count <- as.data.frame(table(employee$Attrition))
colnames(attrition_count) <- c("Attrition", "Count")
ggplot(attrition_count, aes(x = 2, y = Count, fill = Attrition)) +
  geom_col(width = 1, color = "white") +
  coord_polar(theta = "y") +
  xlim(0.5, 2.5) +
  labs(title = "Employee Attrition") +
  theme_void()
```

Employee Attrition

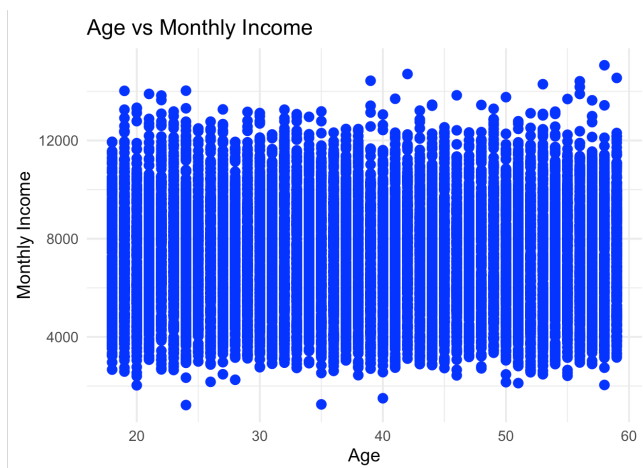


8. Scatter Plot (Age vs Monthly Income)

```
R
library(ggplot2)

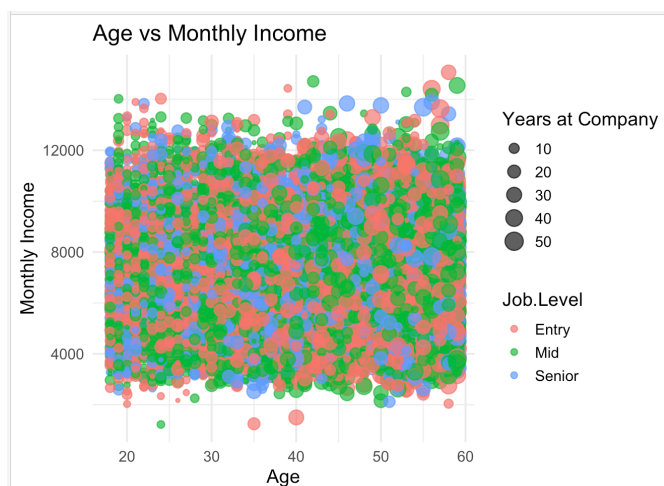
employee <- read.csv("~/Downloads/employee.csv")

ggplot(employee, aes(x = Age, y = Monthly.Income)) +
  geom_point(color = "blue", size = 3) +
  labs(
    title = "Age vs Monthly Income",
    x = "Age",
    y = "Monthly Income"
  ) +
  theme_minimal(base_size = 14)
```



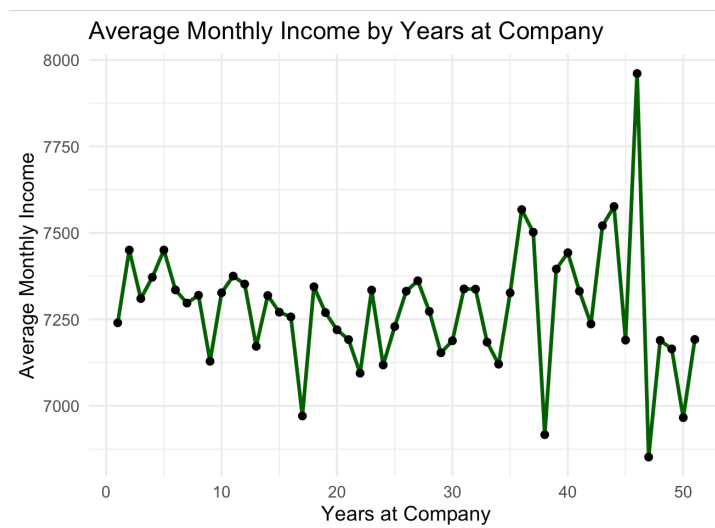
9. Bubble Chart (Age vs Monthly Income)

```
R
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
ggplot(employee,
  aes(x = Age,
    y = Monthly.Income,
    size = Years.at.Company,
    color = Job.Level)) +
  geom_point(alpha = 0.7) +
  labs(
    title = "Age vs Monthly Income",
    x = "Age",
    y = "Monthly Income",
    size = "Years at Company"
  ) +
  theme_minimal(base_size = 14)
```



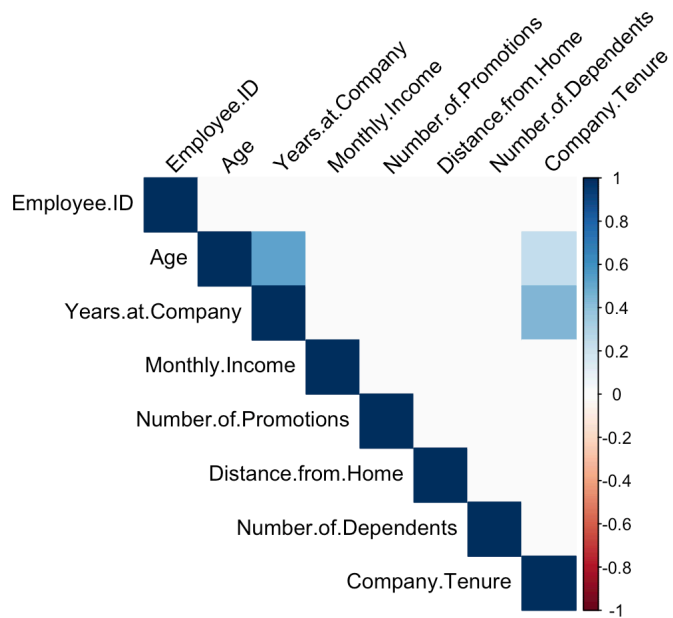
10. Line Chart (Average Income by Years at Company)

```
R
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
income <- aggregate(Monthly.Income ~ Years.at.Company,
                    data = employee,
                    mean)
ggplot(income,
       aes(x = Years.at.Company,
           y = Monthly.Income)) +
  geom_line(color = "darkgreen", linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Average Monthly Income by Years at Company",
    x = "Years at Company",
    y = "Average Monthly Income"
  ) +
  theme_minimal(base_size = 14)
```



11. Correlation Heatmap

```
R
library(corrplot)
employee <- read.csv("~/Downloads/employee.csv")
numeric_data <- employee[sapply(employee, is.numeric)]
cor_matrix <- cor(numeric_data)
corrplot(cor_matrix,
        method = "color",
        type = "upper",
        tl.col = "black",
        tl.srt = 45)
```

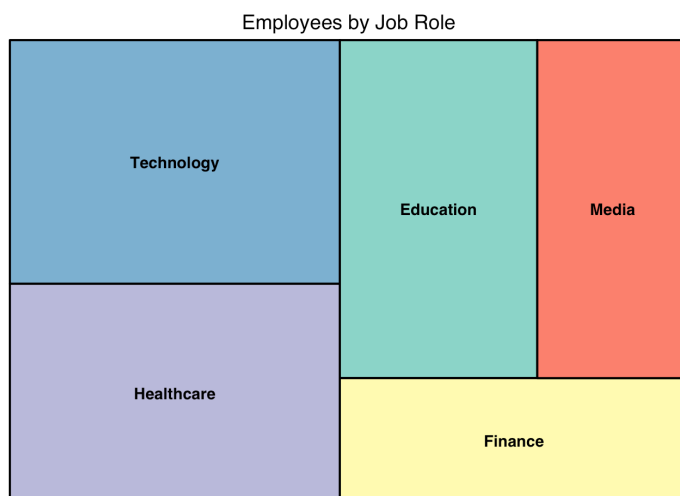


12. Treemap (Employees by Job Role)

```
library(treemap)

employee <- read.csv("~/Downloads/employee.csv")
job_count <- as.data.frame(table(employee$Job.Role))
colnames(job_count) <- c("Job.Role", "Count")

treemap(
  job_count,
  index = "Job.Role",
  vSize = "Count",
  title = "Employees by Job Role",
  palette = "Set3")
```



13. Sunburst Chart (Company Size → Job Level → Attrition)

```
R
library(plotly)
employee <- read.csv("~/Downloads/employee.csv")
plot_ly(
  data = employee,
  type = "sunburst",
  labels = ~Job.Level,
  parents = ~Company.Size,
  values = ~rep(1, nrow(employee)),
  branchvalues = "total"
)
```

14. Heatmap (Categorical Pivot)

This heatmap shows the number of employees for each combination of **Job Satisfaction** and **Performance Rating**.

```
R
library(ggplot2)
employee <- read.csv("~/Downloads/employee.csv")
heat_data <- as.data.frame(table(
  employee$Job.Satisfaction,
  employee$Performance.Rating
))
colnames(heat_data) <- c(
  "Job.Satisfaction",
  "Performance.Rating",
  "Count"
)
ggplot(
  heat_data,
  aes(
    x = Job.Satisfaction,
    y = Performance.Rating,
    fill = Count
  )
) +
  geom_tile(color = "white") +
  geom_text(aes(label = Count),
    color = "black",
    size = 5) +
  scale_fill_gradient(
    low = "lightyellow",
    high = "red"
  ) +
  labs(
    title = "Job Satisfaction vs Performance Rating",
    x = "Job Satisfaction",
    y = "Performance Rating"
  ) +
  theme_minimal(base_size = 14)
```



15. Ridgeline Plot (Monthly Income by Job Role)

```
R
library(ggplot2)
library(ggridges)
employee <- read.csv("~/Downloads/employee.csv")
ggplot(employee,
  aes(x = Monthly.Income,
      y = Job.Role,
      fill = Job.Role)) +
  geom_density_ridges(alpha = 0.7,
    color = "white",
    scale = 1.2) +
  labs(
    title = "Ridgeline Plot of Monthly Income by Job Role",
    x = "Monthly Income",
    y = "Job Role"
  ) +
  theme_ridges() +
  theme(
    legend.position = "none",
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
)
```

